

Notes on Gauge Theory

Bob

Contents

4	Characteristic Classes	2
4.1	Chern-Weil Theory	2
4.2	Examples of Characteristic Classes	5
4.2.1	Pontryagin Classes	5
4.2.2	Euler Classes	7
4.2.3	Chern Classes	7

4 Characteristic Classes

Let $\pi : E \rightarrow M$ be a rank r vector bundle, ∇ be a connection on E . Its curvature $F_\nabla := \nabla^2$ is an $\text{End}(E)$ -valued 2-form on M . Under a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of E , ∇ can be represented by an $r \times r$ matrix of 1-forms ω_α , and F_∇ can be represented by an $r \times r$ matrix of 2-forms Ω_α .

The main goal of this section is the following:

1. An invariant polynomial Φ on $\mathfrak{gl}(r, \mathbb{R})$ determines a global form $\Phi(F_\nabla)$ on M .
2. The global form $\Phi(F_\nabla)$ is closed.
3. The class $[\Phi(F_\nabla)]$ is independent of the choice of ∇ , thus defines an invariant of E .
4. Chern-Weil homomorphism: $\text{Inv}(\mathfrak{gl}(r, \mathbb{R})) \rightarrow H_{\text{dR}}^*(M)$, $\Phi \mapsto [\Phi(F_\nabla)]$.
5. Examples: Chern classes, Pontryagin classes and Euler classes.
6. Generalization to principal G -bundles: $\text{Inv}(\mathfrak{g}) \rightarrow H_{\text{dR}}^*(M)$.

4.1 Chern-Weil Theory

Repeat the settings above: Let $\pi : E \rightarrow M$ be a rank r vector bundle, ∇ a connection on E . The curvature F_∇ is an $\text{End}(E)$ -valued 2-form on M . Under a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of E , ∇ can be represented by an $r \times r$ matrix of 1-forms ω_α , and F_∇ can be represented by an $r \times r$ matrix of 2-forms Ω_α .

Recall that the curvature form Ω_α is defined by

$$\Omega_\alpha = d\omega_\alpha + \omega_\alpha \wedge \omega_\alpha.$$

It satisfies the second Bianchi identity

$$d\Omega_\alpha = \Omega_\alpha \wedge \omega_\alpha - \omega_\alpha \wedge \Omega_\alpha,$$

and the general Bianchi identity

$$d\Omega_\alpha^k = \Omega_\alpha^k \wedge \omega_\alpha - \omega_\alpha \wedge \Omega_\alpha^k.$$

The transformation formula for ω_α and Ω_α is given by

$$\omega_\beta = g_{\alpha\beta}^{-1} \omega_\alpha g_{\alpha\beta} + g_{\alpha\beta}^{-1} d g_{\alpha\beta}, \quad \Omega_\beta = g_{\alpha\beta}^{-1} \Omega_\alpha g_{\alpha\beta}.$$

where $g_{\alpha\beta}$ is the transition function.

Definition 4.1. Let $X = (x_j^i)$ be an $r \times r$ matrix with indeterminate entries. A polynomial $\Phi(X)$ on $\mathfrak{gl}(r, \mathbb{R}) = \mathbb{R}^{r \times r}$ is a polynomial in the variables x_j^i . A polynomial $\Phi(X)$ on $\mathfrak{gl}(r, \mathbb{R})$ is **Ad-invariant (or simply invariant)** if

$$\Phi(A^{-1}XA) = \Phi(X), \quad \forall A \in \text{GL}(r, \mathbb{R}). \quad (4.1)$$

Remark. If X is a matrix of real numbers, then $\Phi(X)$ is a real number. If X is a matrix of 2-forms, then $\Phi(X)$ is also a differential form.

Fact 1. If (4.1) holds for all $r \times r$ matrices X of real numbers, then it also holds when X is a matrix of indeterminate. Thus $\text{Tr}(X)$ and $\det(X)$ are invariant polynomials.

Example 4.2. Let $\det(\lambda I + X) = \lambda^r + f_1(X)\lambda^{r-1} + \dots + f_r(X)$, then $f_k(X)$ is an invariant polynomial of degree k . In fact, $(-1)^k f_k(X)$ is the k -th coefficient of the characteristic polynomial of X .

Example 4.3. Let $\Sigma_k(X) = \text{Tr}(X^k)$ be the trace of X^k , then $\Sigma_k(X)$ is an invariant polynomial of degree k .

Proposition 4.4. Let $\Phi(X)$ be an invariant polynomial on $\mathfrak{gl}(r, \mathbb{R})$. Then $\Phi(\Omega_\alpha)$ and $\Phi(\Omega_\beta)$ coincide on $U_\alpha \cap U_\beta$, *thus determine a global form $\Phi(F_\nabla)$ on M .*

Proof. Since $\Omega_\beta = g_{\alpha\beta}^{-1} \Omega_\alpha g_{\alpha\beta}$, we have

$$\Phi(\Omega_\beta) = \Phi(g_{\alpha\beta}^{-1} \Omega_\alpha g_{\alpha\beta}) = \Phi(\Omega_\alpha), \quad \text{on } U_\alpha \cap U_\beta. \quad \square$$

Proposition 4.5. *The global form $\Phi(F_\nabla)$ is closed.*

To prove this proposition, we need the following fact about invariant polynomials.

Fact 2. The algebra of invariant polynomials on $\mathfrak{gl}(r, \mathbb{R})$ is the polynomial algebra generated by $\Sigma_1, \dots, \Sigma_r$ (or equivalently by $f_1, \dots, f_r$), i.e.,

$$\text{Inv}(\mathfrak{gl}(r, \mathbb{R})) = \mathbb{R}[\Sigma_1, \dots, \Sigma_r] = \mathbb{R}[f_1, \dots, f_r].$$

For example, suppose Φ is a homogeneous invariant polynomial of degree 3 on $\mathfrak{gl}(3, \mathbb{R})$, then Φ must be a linear combination of Σ_1^3 , $\Sigma_1 \Sigma_2$ and Σ_3 , i.e.,

$$\Phi(X) = a \Sigma_1(X)^3 + b \Sigma_1(X) \Sigma_2(X) + c \Sigma_3(X), \quad a, b, c \in \mathbb{R}.$$

Proof of Proposition 4.5. Since Φ is a polynomial in $\Sigma_1, \dots, \Sigma_r$, it suffices to show that $d\text{Tr}(F_\nabla) = 0$ for all k . We have

$$d\text{Tr}(\Omega_\alpha^k) = \text{Tr}(d\Omega_\alpha^k) = \text{Tr}(\Omega_\alpha^k \wedge \omega_\alpha - \omega_\alpha \wedge \Omega_\alpha^k) = 0. \quad \square$$

Proposition 4.6. *The class $[\Phi(F_\nabla)]$ is independent of the choice of ∇ . It is a (differential) topological invariant of E .*

Proof. For the same reason as above, it suffices to show that $[\text{Tr}(F_\nabla)]$ is independent of the choice of ∇ . Let ∇_0, ∇_1 be two connections on E , then $\nabla_t = (1-t)\nabla_0 + t\nabla_1$ is a family of connections on E , whose connection form is given by

$$\omega_t = (1-t)\omega_0 + t\omega_1,$$

and curvature form is

$$\Omega_t = d\omega_t + \omega_t \wedge \omega_t.$$

So

$$\frac{d\Omega_t}{dt} = \frac{d}{dt}(d\omega_t + \omega_t \wedge \omega_t) = \frac{d}{dt}(d\omega_t) + \frac{d\omega_t}{dt} \wedge \omega_t + \omega_t \wedge \frac{d\omega_t}{dt} = d \frac{d\omega_t}{dt}$$

Then

$$\begin{aligned}
\frac{d}{dt} \text{Tr}(\Omega_t^k) &= \text{Tr} \left(\frac{d}{dt} \Omega_t^k \right) = \text{Tr} \left(\frac{d\Omega_t}{dt} \wedge \Omega_t^{k-1} + \cdots + \Omega_t^{k-1} \wedge \frac{d\Omega_t}{dt} \right) \\
&= k \text{Tr} \left(\frac{d\Omega_t}{dt} \wedge \Omega_t^{k-1} \right) = k \text{Tr} \left(\left(d \frac{d\omega_t}{dt} \right) \wedge \Omega_t^{k-1} \right) \\
&= k \text{Tr} \left(d \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right) \right) = d \, k \text{Tr} \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right).
\end{aligned}$$

Therefore,

$$\begin{aligned}
\text{Tr}(\Omega_1^k) - \text{Tr}(\Omega_0^k) &= \int_0^1 \frac{d}{dt} \text{Tr}(\Omega_t^k) dt = \int_0^1 d \, k \text{Tr} \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right) dt \\
&= d \int_0^1 k \text{Tr} \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right) dt.
\end{aligned} \tag{4.2}$$

Notice that

$$\omega_{t,\beta} = g_{\alpha\beta}^{-1} \omega_{t,\alpha} g_{\alpha\beta} + g_{\alpha\beta}^{-1} d g_{\alpha\beta} \implies \frac{d\omega_{t,\beta}}{dt} = g_{\alpha\beta}^{-1} \frac{d\omega_{t,\alpha}}{dt} g_{\alpha\beta}, \quad \Omega_{t,\beta} = g_{\alpha\beta}^{-1} \Omega_{t,\alpha} g_{\alpha\beta}.$$

So $\text{Tr} \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right)$ defines a global form on M . By (4.2), we have $[\text{Tr}(\Omega_1^k)] = [\text{Tr}(\Omega_0^k)]$.

For a general invariant polynomial Φ , we can express Φ as a linear combination of monomials $\Sigma_1^{k_1} \cdots \Sigma_r^{k_r}$. Then $\Phi(\Omega_t)$ is a linear combination of $\Sigma_1^{k_1}(\Omega_t) \wedge \cdots \wedge \Sigma_r^{k_r}(\Omega_t)$. Thus $[\Phi(\Omega_1)] = [\Phi(\Omega_t)] = [\Phi(\Omega_0)]$. $\square$

Definition 4.7. By above propositions, we define the **Chern-Weil homomorphism** as

$$\begin{aligned}
c_E : \text{Inv}(\mathfrak{gl}(r, \mathbb{R})) &\rightarrow H_{\text{dR}}^*(M), \\
\Phi &\mapsto [\Phi(F_\nabla)].
\end{aligned}$$

This map sends $\Phi(X)\Psi(X)$ to $[\Phi(F_\nabla) \wedge \Psi(F_\nabla)] = [\Phi(F_\nabla)] \cdot [\Psi(F_\nabla)]$, thus is an algebra homomorphism. This map does not depend on the choice of ∇ .

Definition 4.8. A **characteristic class** of a vector bundle $\pi : E \rightarrow M$ is a map

$$c : \text{Vect}_k(M) \rightarrow H_{\text{dR}}^*(M), \quad E \mapsto c(E),$$

such that if $f : N \rightarrow M$ is a smooth map, then

$$c(f^*E) = f^*c(E).$$

In functional language, a **characteristic class** is a natural transformation from the functor $\text{Vect}_k(-)$ to $H_{\text{dR}}^*(-)$. The naturality condition can be visualized by the following commutative diagram:

$$\begin{array}{ccc}
\text{Vect}_k(M) & \xrightarrow{c} & H_{\text{dR}}^*(M) \\
f^* \uparrow & & \uparrow f^* \\
\text{Vect}_k(N) & \xrightarrow{c} & H_{\text{dR}}^*(N)
\end{array}$$

Naturality of Chern-Weil homomorphism

Suppose $\pi : E \rightarrow M$ is a rank r vector bundle, ∇ a connection on E . Let $f : N \rightarrow M$ be a smooth map, it induces a pullback bundle $f^*E \rightarrow N$ and a pullback connection $f^*\nabla$ on f^*E

$$(f^*\nabla)(f^*s) := f^*(\nabla s), \quad \forall s \in \Gamma(E).$$

Suppose ω is the connection form of ∇ , then $f^*\omega$ is the connection form of $f^*\nabla$. The curvature form of $f^*\nabla$ is therefore $f^*(d\omega + \omega \wedge \omega) = f^*\Omega$. For any invariant polynomial Φ , we have

$$\Phi(f^*\Omega) = f^*\Phi(\Omega) \implies c_{f^*E}(\Phi) = [\Phi(f^*\Omega)] = f^*[\Phi(\Omega)] = f^*c_E(\Phi).$$

which shows the naturality of Chern-Weil homomorphism.

4.2 Examples of Characteristic Classes

We have introduced a powerful machinery to construct characteristic classes of vector bundles. In this subsection, we will give a few examples of characteristic classes.

4.2.1 Pontryagin Classes

In this subsection, we only consider real vector bundles. Recall that

$$\det(\lambda I + X) = \lambda^r + f_1(X)\lambda^{r-1} + \cdots + f_r(X),$$

in particular, $f_1(X) = \text{Tr}(X)$. If X is a skew-symmetric matrix, then $f_1(X) = 0$. In general we have the following proposition.

Proposition 4.9. *If Ω is skew-symmetric, then $\text{Tr}(\Omega^k) = 0$ for all odd k . Moreover, $\Phi(\Omega) = 0$ for all invariant polynomials Φ of odd degree.*

Proof. Since Ω is skew-symmetric,

$$(\Omega \wedge \Omega)^T = \Omega^T \wedge \Omega^T = (-\Omega) \wedge (-\Omega) = \Omega \wedge \Omega,$$

and

$$(\Omega \wedge \Omega \wedge \Omega)^T = (\Omega \wedge \Omega)^T \wedge \Omega^T = \Omega \wedge \Omega \wedge (-\Omega) = -\Omega \wedge \Omega \wedge \Omega.$$

By induction, we have Ω^k is skew-symmetric for odd k , thus $\text{Tr}(\Omega^k) = 0$.

For a general invariant polynomial Φ of odd degree, since Φ is a linear combination of monomials $\Sigma_1^{k_1} \cdots \Sigma_r^{k_r}$ with $k_1 + 2k_2 + \cdots + rk_r$ odd, at least one of $k_1, k_3, \dots$ is odd, thus $\Phi(\Omega)$ must be zero. $\square$

Since $[\Phi(F_\nabla)]$ is independent of the choice of ∇ , we can choose a special connection ∇ such that F_∇ is skew-symmetric. To do this, we choose a Riemannian metric on E , then there exists connections on E that are compatible with the metric. Under a local orthonormal frame, the connection form ω is skew-symmetric, so as the curvature form Ω . By the above proposition, $[\Phi(F_\nabla)] = 0$ for all invariant polynomials Φ of odd degree.

Corollary 4.10. *For a real vector bundle E , the ring of characteristic classes of E has two sets of generators:*

$$[\mathrm{Tr}(\Omega^2)], [\mathrm{Tr}(\Omega^4)], [\mathrm{Tr}(\Omega^6)], \dots;$$

and

$$[f_2(\Omega)], [f_4(\Omega)], [f_6(\Omega)], \dots.$$

Definition 4.11. The k -th **Pontryagin class** of a real vector bundle E is defined to be

$$p_k(E) := (-1)^k \left[f_{2k} \left(\frac{1}{2\pi} F_\nabla \right) \right] = \left[f_{2k} \left(\frac{i}{2\pi} F_\nabla \right) \right] \in H^{4k}(M).$$

Remark. The factor i is for the sign convention. The factor $\frac{1}{2\pi}$ is for the integrality of Pontryagin classes. A class $\alpha \in H^q(M)$ is called **integral** if $\int_S \alpha \in \mathbb{Z}$ for all q -dimensional closed oriented submanifolds S of M .

By definition,

$$\begin{aligned} \det \left(\lambda I + \frac{i}{2\pi} F_\nabla \right) &= \lambda^r + f_1 \left(\frac{i}{2\pi} F_\nabla \right) \lambda^{r-1} + \dots + f_r \left(\frac{i}{2\pi} F_\nabla \right) \\ &= \lambda^r + p_1(E) \lambda^{r-2} + \dots + p_{[r/2]}(E) \lambda^{r-2[r/2]}. \end{aligned}$$

Apply $\lambda = 1$,

$$\det \left(I + \frac{i}{2\pi} F_\nabla \right) = 1 + p_1(E) + \dots + p_{[r/2]}(E) =: p(E).$$

We call $p(E)$ the **total Pontryagin class** of E .

Proposition 4.12. *For two real vector bundles E and E' , we have*

$$p(E \oplus E') = p(E) \cdot p(E').$$

Proof. Let ∇ and ∇' be connections on E and E' , then $\nabla \oplus \nabla'$ is a connection on $E \oplus E'$. Under a suitable local trivialization, the curvature form of $\nabla \oplus \nabla'$ is given by

$$F_{\nabla \oplus \nabla'} = \begin{pmatrix} F_\nabla & 0 \\ 0 & F_{\nabla'} \end{pmatrix} \implies \Omega_{E \oplus E'} = \begin{pmatrix} \Omega & 0 \\ 0 & \Omega' \end{pmatrix}.$$

So

$$\begin{aligned} p(E \oplus E') &= \det \left(I + \frac{i}{2\pi} \Omega_{E \oplus E'} \right) \\ &= \det \begin{pmatrix} I + \frac{i}{2\pi} \Omega & \\ & I + \frac{i}{2\pi} \Omega' \end{pmatrix} \\ &= \det \left(I + \frac{i}{2\pi} \Omega \right) \cdot \det \left(I + \frac{i}{2\pi} \Omega' \right) \\ &= p(E) \cdot p(E'). \end{aligned}$$

□

4.2.2 Euler Classes

If in addition E is orientable and we choose an orientation of E . Then we can make the local orthonormal frame $e = [e_1, \dots, e_r]$ of E to be compatible with the orientation, so the transition function between two frames is in $\mathrm{SO}(r)$ and the connection form ω is skew-symmetric. This inspires us to consider the $\mathrm{SO}(r)$ -invariant polynomial on $\mathfrak{so}(r)$.

Fact 3. For r odd, $\mathrm{Inv}(\mathfrak{so}(r))$ is generated by $\Sigma_1, \dots, \Sigma_r$ (or equivalently by $f_1, \dots, f_r$). For r even, $\mathrm{Inv}(\mathfrak{so}(r))$ has one more generator Pf , the **Pfaffian**.

The Pfaffian of skew-symmetric matrices

4.2.3 Chern Classes